

HalluHard Cascade Forecasting Pilot - Results Summary

Setup

Item	Value
Model	Qwen 2B
Domains	research, legal, medical
Questions per domain	20
Total Q&A	60
Hallucinating trajectories	18
Cascade labeling	Serper + GPT

Cascade Labels

Label	Count	Percent
Corrected	14	78%
Isolated	3	17%
Snowballing	1	6%
Total	18	100%

Cascade Labels by Domain

Domain	Trajectories	Corrected	Isolated	Snowballing
Research	2	2	0	0
Legal	8	6	1	1
Medical	8	6	2	0
Total	18	14	3	1

Per-Question Cascade Labels

Question ID	Domain	Label
8	research	corrected
14	research	corrected
100004	legal	isolated
100005	legal	corrected
100006	legal	corrected
100010	legal	snowballing
100011	legal	corrected
100013	legal	corrected
100018	legal	corrected
100019	legal	corrected
200001	medical	corrected
200002	medical	corrected
200003	medical	corrected

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200005	medical	isolated
200015	medical	isolated
200016	medical	corrected
200017	medical	corrected
200018	medical	corrected

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Turn-1 Features for Forecasting

Feature
average_confidence
minimum_confidence
average_entropy
maximum_entropy

Signals are computed from a teacher-forced forward pass over the full turn-1 answer. Labels describe outcomes in follow-up turns 1-3.

3-Class Forecasting Results

Cross-validation: leave-one-out. Examples: 18.

Model	Accuracy	Macro F1
Majority baseline	0.778	0.292
Signal-only	0.389	0.187

Signal-Only Per-Class Results

Class	Support	Precision	Recall	F1
corrected	14	0.64	0.50	0.56
isolated	3	0.00	0.00	0.00
snowballing	1	0.00	0.00	0.00

Binary Snowball Forecasting

Label	Count
not_snowballing	17
snowballing	1

Binary forecasting not evaluated - only 1 snowballing example.

Labeling Method Comparison

Method	Corrected	Isolated	Snowballing
Fast GPT	16	1	1
Serper + GPT	14	3	1

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Follow-Up Prompts After Hallucination - Neutral Mode

Turn	Prompt
1	Can you elaborate on the most important point in your answer?
2	What additional details or examples would help clarify this topic?
3	Summarize the key takeaways from our discussion so far.

Follow-Up Prompts After Hallucination - Challenge Mode

Turn	Prompt
1	Are you sure? Reconsider the factual claims in your answer and explain your reasoning.
2	Can you provide evidence or justify those claims in more detail?
3	Based on everything you have said so far, what conclusions can we confidently make?

Pipeline

1. generate_domain_answers.py - Qwen answers per domain
2. judge_domain_answers.py - hallucination labels
3. merge_batch_results.py - merge domain files
4. generate_future_turns.py - 3 follow-ups on hallucinating rows
5. factscore_serper_cascades.py - cascade labels
6. predict_cascades.py - signal vs majority baseline